higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

AUGUST EXAMINATION

INDUSTRIAL ELECTRONICS N5

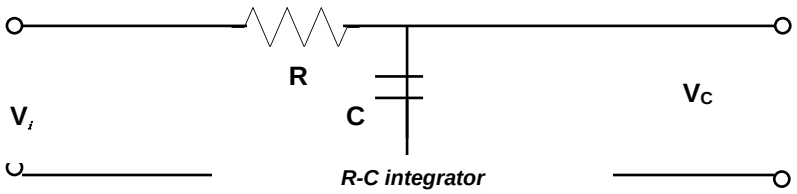
27 JULY 2015

This marking guideline consists of 11 pages.

Copyright reserved

QUESTION 1: ALTERNATING- CURRENT THEORY

1.1

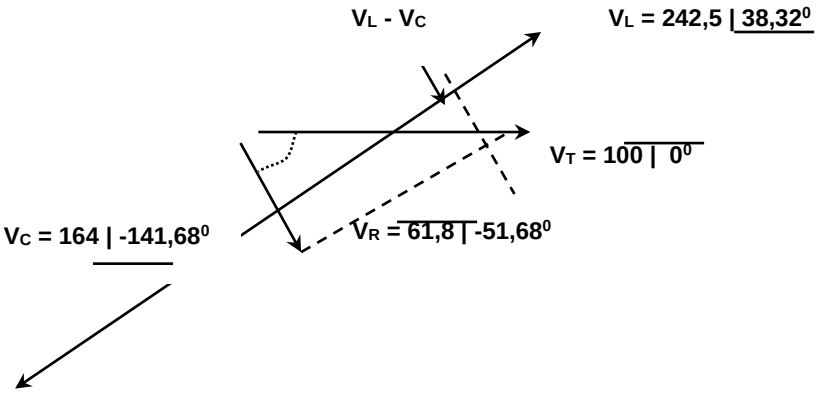


Low-pass filter (3)

1.2 1.2.1 Greater

1.2.2 Square waveforms (2 x 1) (2)

1.3 1.3.1



(3)

1.3.2 A $X_L = \frac{V_L}{I_T}$

$= \frac{242,5 \angle 38,32^0}{5,15 \angle -51,68^0}$

$= 47,087 \angle 90^0$ Ω ⟶ (2)

B $X_C = \frac{V_C}{I_T}$

$$= \frac{164 \angle -141{,}68^0}{5{,}15 \angle -51{,}68^0}$$

$$= 31{,}845 \angle -90^0\ \Omega \longrightarrow$$

(2)

C $R = \frac{V_R}{I_T}$

$$= \frac{61{,}8 \angle -51{,}68^0}{5{,}15 \angle -51{,}68^0}$$

$$= 12 \angle 0^0\ \Omega \longrightarrow$$

(2)
[14]

QUESTION 2: POWER SUPPLIES

2.1 2.1.1

$$V'_{DC} = \frac{R_L}{R + R_L} \times V_{DC}$$

$$\therefore V_{DC} = \frac{R + R_L}{R_L} \times V'_{DC}$$

$$= \frac{10 + 300}{300} \times 12$$

$$= 12{,}4\ \mathbf{V} \longrightarrow$$

(3)

2.1.2

$$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} \times V_{r(rms)}$$

$$\therefore V_{r(rms)} = \frac{\sqrt{R^2 + X_C^2}}{X_C} \times V'_{r(rms)}$$

$$= \frac{\sqrt{10^2 + 8^2}}{8} \times 0{,}8$$

$$= 1{,}281 \text{ V} \longrightarrow \quad (3)$$

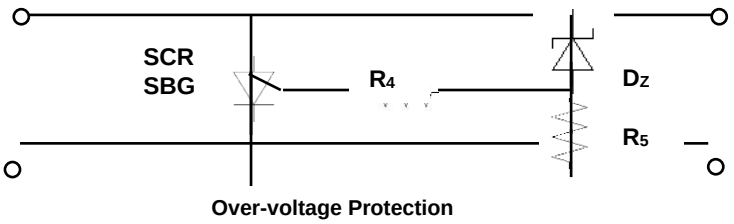
2.1.3

$$r' = \frac{V'_{r(rms)}}{V'_{DC}}$$

$$= \frac{0{,}8}{12}$$

$$= 0{,}067 \longrightarrow \quad (2)$$

2.2



Over-voltage Protection

Under normal output voltages, voltage across R5 will be low thus SCR will be off. When maximum allowed voltage is exceeded voltage drop across R5 increases and SCR is switched on. The SCR causes a short circuit and the output voltage drops to zero. (6)

[14]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1

3.1.1 $V_e = \frac{1}{10} \times V_{CC}$

$= \frac{12}{10}$

$= 1,2$ **V**

$R_e = \frac{V_e}{I_c}$

$= \frac{1,2}{20 \times 10^{-3}}$

$= 60\ \Omega$(3) → (3)

3.1.2 $C_e = \frac{10}{2\pi f R_e}$

$= \frac{10}{2 \times \pi \times 100 \times 60}$

$= 265,26\ \mu f$ → (2)

3.1.3 $R_C = \frac{V_{cc} - V_{ce} - V_e}{I_c}$

$= \frac{12 - 6 - 1,2}{20 \times 10^{-3}}$

$= 240\ \Omega$ → (2)

3.1.4

$$I_B = \frac{I_C}{I_B}$$

$$= \frac{20\times10^{-3}}{250}$$

$$= 80\ \mu A$$

$$R_B = \frac{V_{cc} - V_e - V_{be}}{I_B}$$

$$= \frac{12-1{,}2-0{,}6}{80\times10^{-6}}$$

$$= 127{,}5\ k\Omega \longrightarrow \quad (3)$$

3.2 3.2.1

$$Z_L = \frac{R_C \times R_L}{R_C + R_L}$$

$$= \frac{240\times10000}{240+10000}$$

$$= 234{,}375\ \Omega$$

$$A_i = \frac{h_{fe}}{1+h_{oe}.Z_L}$$

$$= \frac{250}{1 + 20\times10^{-6}\times234{,}375}$$

$$= 248{,}834 \longrightarrow \quad (4)$$

3.2.2

$$A_v = \frac{-h_{fe}.Z_L}{h_{ie} + \left(h_{ie}h_{oe} - h_{fe}h_{re}\right)Z_L}$$

$$= \frac{-250\times234{,}375}{1300 + \left(1300\times20\times10^{-6} - 250\times2\times10^{-4}\right)\times234{,}375}$$

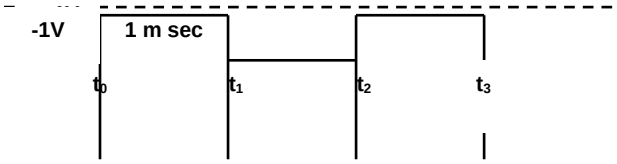
$$= 45{,}268 \longrightarrow$$

(3)
[17]

Copyright reserved

QUESTION 4: OPERATIONAL AMPLIFIERS

4.1



$t = \frac{1}{f}$ = $\frac{1}{500}$ = 2 m sec

$\therefore 360^0$ = 2 m sec thus 180^0 = 1 m sec

$V_{O(t_1)}$ = $-\frac{1}{RC}\int_{t0}^{t1} V dt$ + initial condition of capacitor

= $-\frac{1}{RC}\int_{10}^{t1} V_i(t_1)$ + $V_C(t_0)$ but $t_0 = 0$

= $\left(\frac{1}{10\times10^{6}\times0{,}01\times10^{-6}}\right)\int_{t0}^{t1}(1)(0{,}001)$ + 0

= $-\frac{1}{0{,}01}\int_{t0}^{t1}0{,}001$

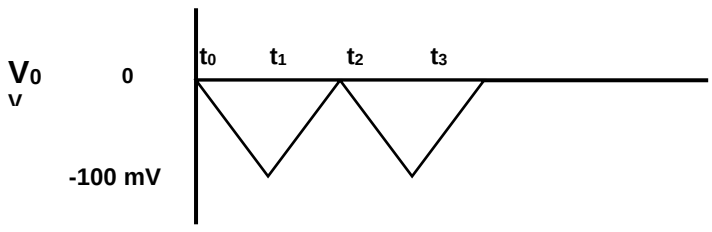
= $-0{,}1\,V$ or $-100\ mV$

$V_{O(t_1)}$ = $-\frac{1}{RC}\int_{11}^{t2} V_i(t_2)$ + $V_C(t_1)$

= $(-100)\int_{t1}^{t2}(-1)(0{,}001)$ + $(-0{,}1)$

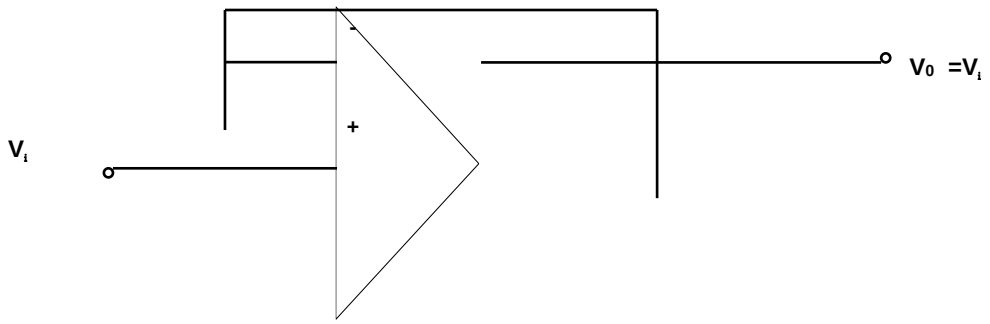
= + 0,1 - 0,1

= 0 V



(9)

4.2



The voltage follower

CIRCUIT FUNCTIONS

a) Acts as buffer between stages

b) Used for impedance matching

c) Power amplification

(4)
[13]

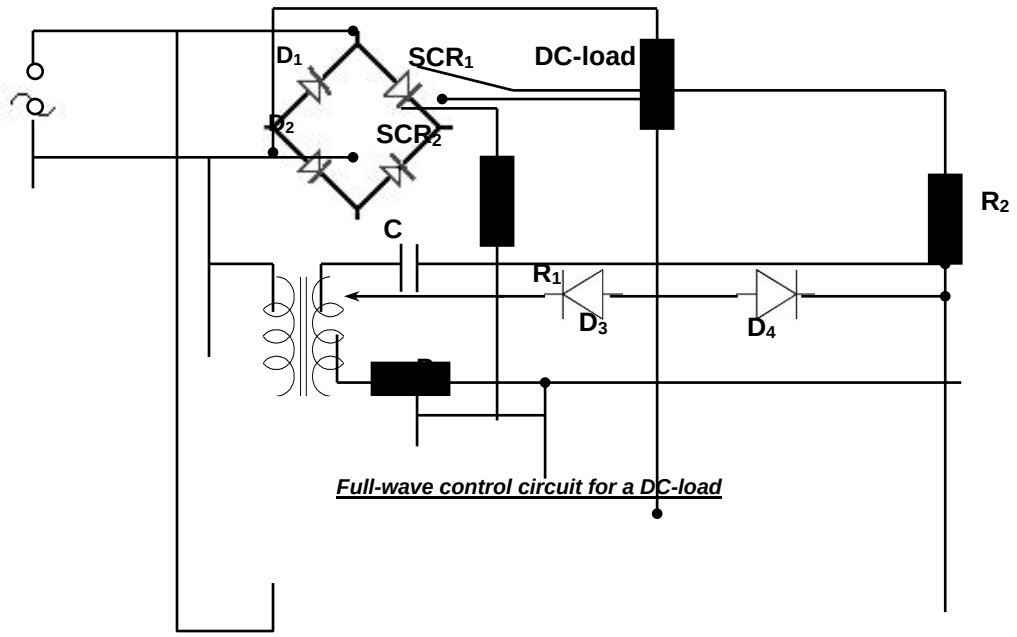
QUESTION 5: INTEGRATED CIRCUITS

5.1 All unutilised inputs must be connected to V_{DD} or V_{SS} (2)

5.2 5.2.1 True
5.2.2 False
5.2.3 True

(3 x 1) (3)
[5]

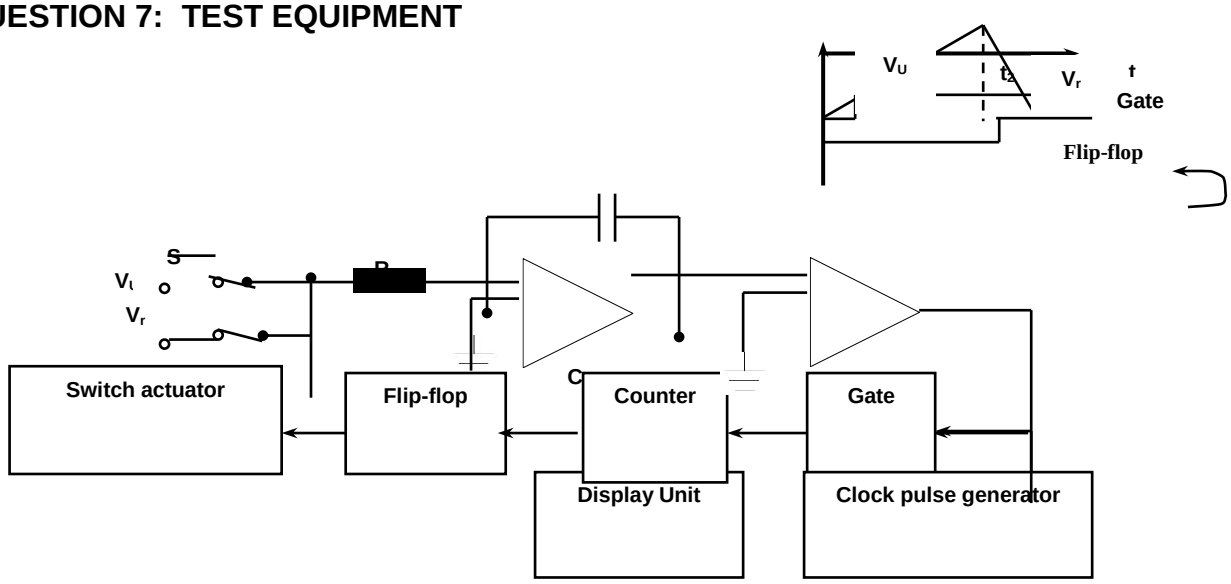
QUESTION 6: ELECTRONIC PHASE CONTROL



Full-wave control circuit for a DC-load

[6]

QUESTION 7: TEST EQUIPMENT



Double-slope voltmeter

[6]

QUESTION 8: OSCILLATORS

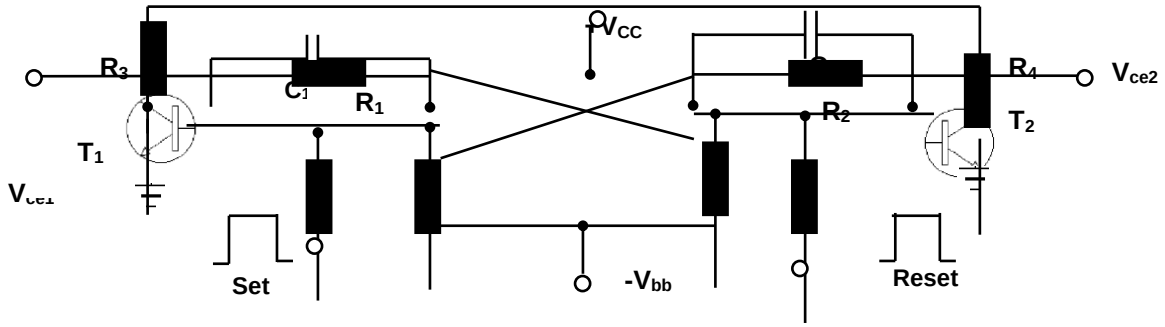
8.1 8.1.1 Two semi-stable conditions

8.1.2 One stable- and one semi-stable condition

8.1.3 Two stable conditions

(3 x 1) (3)

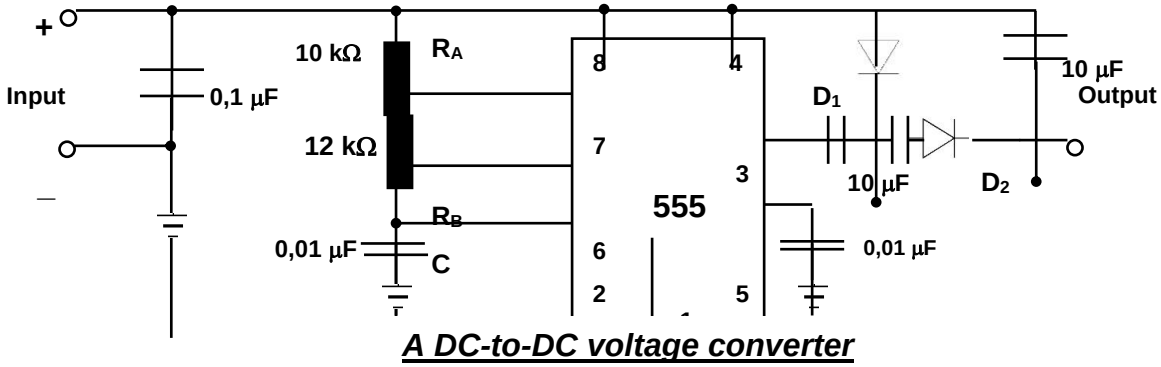
8.2



The bi-stable multi-vibrator

(4)

8.3

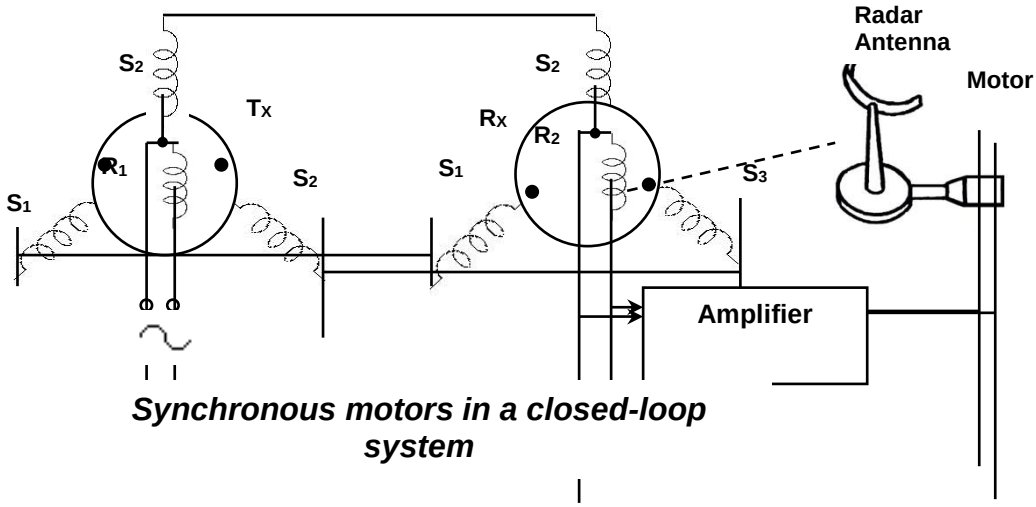


A DC-to-DC voltage converter

(5)
[12]

QUESTION 9: TRANSDUCERS

9.1

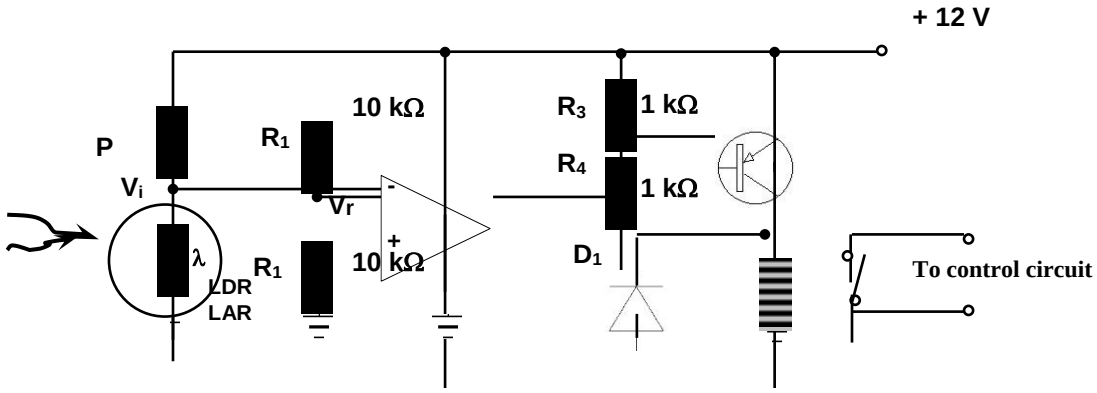


(4)

Synchronous motors in a closed-loop system

9.2 The transmitter T_x and receiver R_x are in balance as soon as the rotors stand at the same angle. If the rotor of the transmitter turns at an angle θ, the rotor of the receiver will produce an error signal to turn the motor. The antenna also turns the rotor of the receiver at the same angle θ so that the error signal becomes zero. (4)

9.3



Precision light switch

(5)
[13]

TOTAL: 100

Copyright reserved